

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

B.S. in Computer Science, Systems Concentration

May 2027 | MGPA: 3.6/4.0

Boston, MA

Coursework: Computer Systems, Computer Networks, Digital Design & Computer Architecture, Foundations of Software Engineering, Algorithms & Data Structures, Object-Oriented Design, Systems Security, Cybersecurity, Foundations of Data Science

EXPERIENCE

Wayfair

Software Engineering Co-op

Jul 2026 – Present

Boston, MA

- Own front-of-house device stack – e-label hardware fleet, mobile, and customer/associate-facing POS – across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold)
- Building a supplier-agnostic e-label device API that unifies 4 hardware suppliers behind one backend interface, powering new outlet launches
- Shipped POS checkout flows end-to-end across the SDLC: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Software Engineer

Jul 2025 – Jun 2026

Boston, MA

- Overhauled Relay JMS message-broker serialization with json-io across 20+ backend microservices, eliminating timeouts at scale in a high-throughput message-passing system
- Built Java Spring backend microservice for CME credit issuance, owning design through release independently
- Optimized backend search and filtering queries across 20+ microservices, improving query relevance at scale
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Software Architect

Jan 2025 – Aug 2026

New York, NY

- Architected Flask backend and retrieval pipeline, selecting methodologies independently under general direction
- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Evolved backend retrieval stack from LangChain to Vespa to zero-ops Bedrock KB, cutting operational overhead
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Continue as technical advisor on backend architecture and agentic-workflow direction

Chevron New Energies

Software Engineer Intern

May 2024 – Aug 2024

Houston, TX

- Built RESTful APIs and sensor-to-cloud data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a fault-tolerant, linearizable distributed key-value store in Rust with from-scratch Raft consensus, async message passing over Tokio, and an LSM-tree storage engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving model detection
- Designed AWS Neuron SDK deployment path for onboard low-power inference under hard real-time limits

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector retrieval over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

SKILLS & HONORS

Languages/Tools: Rust, C, C++, Java, Python, Bash, Linux, PostgreSQL, GraphQL, TypeScript, JavaScript, Groovy, Docker, Kubernetes, Terraform, Jenkins, Buildkite, AWS, Azure

Frameworks & Libraries: Flask, FastAPI, Node.js, Express, React, Next.js, PyTorch, TensorFlow, Pandas

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor